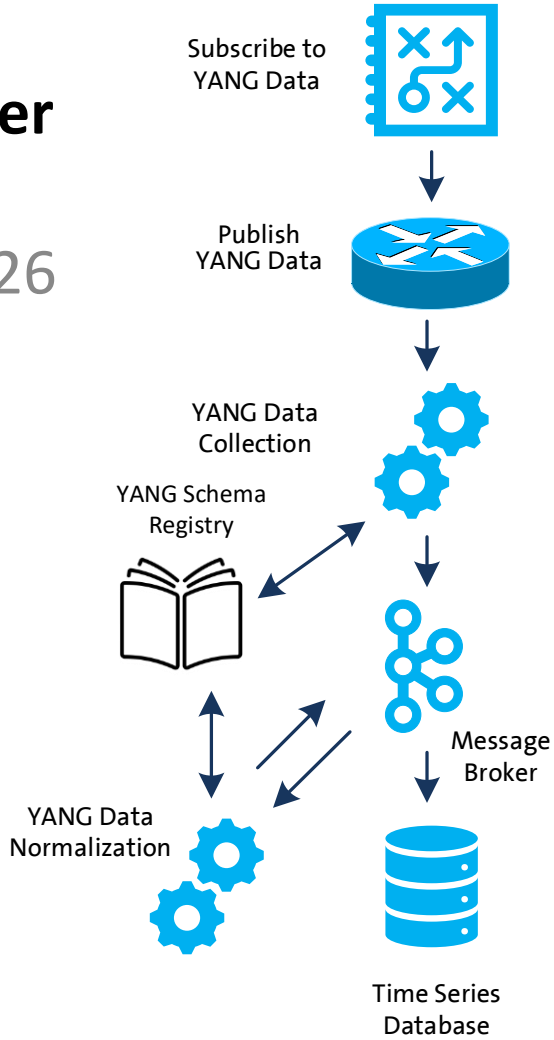
Validate YANG-Push to Message Broker End-To-End Data Processing Chain

IETF 125 Hackathon, March 14-15th 2026



NMOP & NETCONF WG

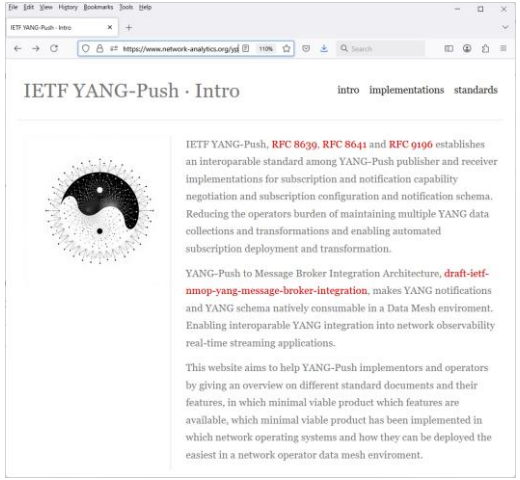
Hackathon Plan, Software and Website

Test Plan

Validate and verify

- 5 YANG-Push Publishers
- 2 YANG-Push Receivers
- 2 YANG-Push Network Telemetry Message
- 1 YANG Message Broker Producer and Schema Registry
- 3 YANG Message Broker Consumers

implementation in YANG data integration automation. Subscribe to YANG data on YANG-Publisher, obtain and register all YANG modules necessary to build YANG schema tree, register YANG schemas to Schema Registry and verify YANG notifications against scheme trees and produce and consume from Message Broker.



Software

https://www.network-analytics.org/yp/how-to-deploy.html

- YANG-Push Publisher - Cisco IOS XR, 6WIND VSR, Huawei NE (Router) and MA (OLT) and Arrcus Arcos
- YANG-Push Receiver – Netgauze and Pmacct
- YANG Message Broker Producer – Netgauze
- YANG Message Broker Consumer – Netgauze, Ciena Blueplanet UAA Workflow Engine, Cisco Crosswork Ingestion Service
- YANG Schema Registry – Apache Kafka YANG Plugin
- Libyang - https://github.com/CESNET/libyang/compare/master...devel
- Yangkit - https://github.com/yang-central/yangkit/pull/41
- Wireshark - udp-notif dissector

Hackathon – Repositories

Test Results on End to End Tests

- Netconf RPCs and YANG-Push JSON encoded notifications with wire captures from receiver
- YANG-Push Receiver cache content generation with YANG library context, YANG modules, metadata and notification schema validation logs
- YANG Message Broker Producer serialization logs
- YANG Schema Registry schema examples
- YANG Message Broker Consumer, serialization logs, YANG library context, YANG modules and message schema validation

ietf-network-analytics-document-status / 125 / Hackathon / Add file ···

graf3net Cisco Crosswork Ingestion Service d53f11b · last month History

Name	Last commit message	Last commit date
..		
messagebroker-consumer	Cisco Crosswork Ingestion Service	last month
messagebroker-producer/netgauze-prod...	testcases updated	last month
yangpush-publisher	testcases updated	last month
yangpush-receiver/netgauze-receiver	nit	last month
how-to-install-netgauze.md	how to install netgauze	last month
readme.md	updated	last month

readme.md

YANG-Push and Message Broker Architecture

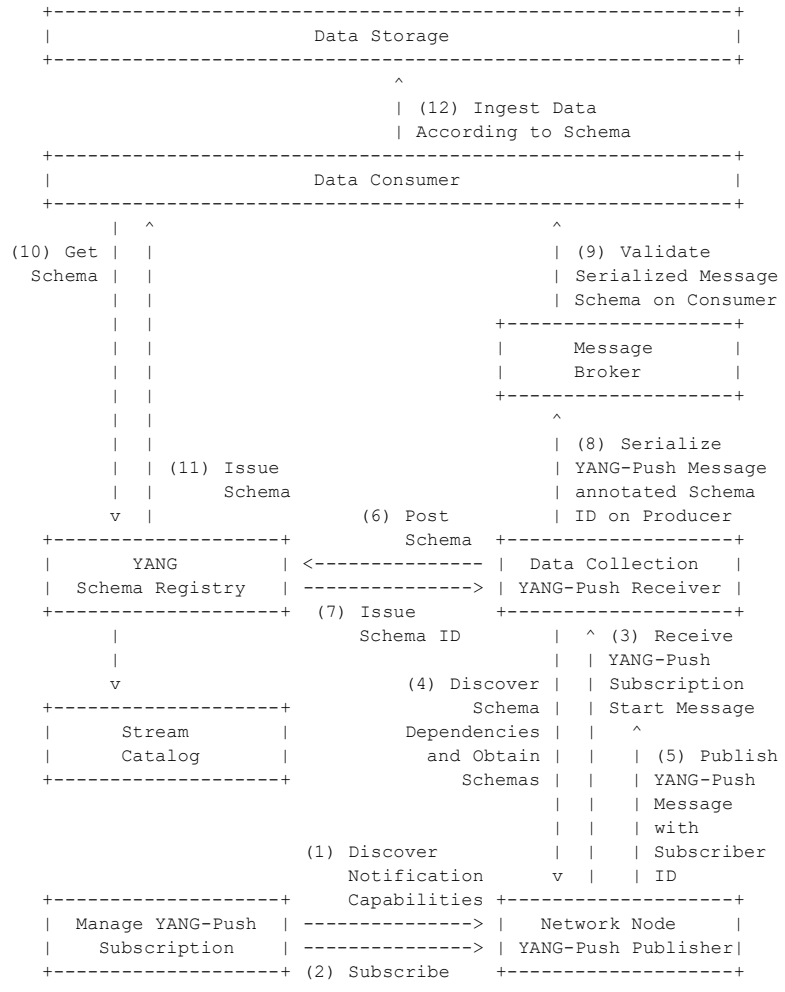
As described in https://datatracker.ietf.org/doc/html/draft-ietf-nmop-yang-message-broker-integration-10#section-4

https://github.com/network-analytics/ietf-network-analytics-document-status/tree/main/125/Hackathon/

An Architecture for YANG-Push to Message Broker Integration

draft-ietf-nmop-yang-message-broker-integration
draft-ietf-nmop-message-broker-telemetry-message
draft-ietf-nmop-yang-message-broker-message-key
draft-netana-nmop-message-broker-bmp-telemetry-message

- Subscription to YANG Notifications
  RFC 8639
- Subscription to YANG Notifications for Datastore Updates
  RFC 8641
- UDP-based Transport for Configured Subscriptions
  draft-ietf-netconf-udp-notif
- Subscription to Distributed Notifications
  draft-ietf-netconf-distributed-notif
- Extensible YANG Model for YANG-Push Notifications
  draft-ietf-netconf-notif-envelope
- Support of Versioning in YANG Notifications Subscription
  draft-ietf-netconf-yang-notifications-versioning
- YANG Modules Describing Capabilities for Systems and Datastore Update Notifications
  RFC 9196
- YANG Notification Transport Capabilities
  draft-ietf-netconf-yp-transport-capabilities
- YANG Library
  RFC 8525
- Augmented-by Addition into the IETF-YANG-Library
  draft-ietf-netconf-yang-library-augmentation
- Validating anydata in YANG Library context
  draft-ietf-netmod-yang-anydata-validation
- Encoding of Data Modeled with YANG in the CBOR
  RFC 9254



YANG-Push

IETF 125 Publisher Implementation Status

	6WIND VSR	Huawei NE	Huawei MA	Cisco IOS XR	Arrcus Arcos	Open-Source
RFC 8639 YANG-Push Subscription	✓	✓	✓	✓	✓	
RFC 8641 YANG-Push Notification	✓	✓	✓	✓	✓	
draft-ietf-netconf-udp-notif	✓	✓	✓	✓	✓	✓
draft-ietf-netconf-distributed-notif	✓	✓	✓		na	
draft-ietf-netconf-notif-envelope	✓	✓	✓	✓	✓	
draft-ietf-netconf-yang-notifications-versioning	✓	✓	✓	✓	✓	
RFC 8525 YANG Library	✓	✓	✓	✓	✓	
draft-ietf-netconf-yang-library-augmentation	✓	✓	✓	✓		✓
RFC 9196 System and Notification Capabilities			✓	P	✓	
draft-netana-netconf-yp-transport-capabilities			✓	✓	✓	
RFC 9254 CBOR Named Identifiers	✓					

[NetGauze](#)
[Pmacct](#)
[udp-notif-c-collector](#)
[yang-library-augmentedby](#)

Green marked describes new capability at IETF 125. "P" to partially implemented.

Operational IETF/IEEE YANG

IETF 125 Module Implementation Status

	6WIND VSR	Huawei NE	Huawei MA	Cisco IOS XR	Arrcus Arcos
ietf-interfaces.yang (inventory, state and statistic)	✓	P	✓		
ietf-hardware.yang (inventory, state and statistic)	(✓)		✓		
ietf-alarms.yang (state)	✓		✓		
ieee802-dot1ab-lldp.yang (inventory and state)	(✓)				
ieee802-dot1ax.yang (inventory, state and statistic)					
ietf-bfd-ip-sh.yang (state)					
ietf-bfd-ip-mh.yang (state)					
ietf-bfd-lag.yang (state)					
ietf-isis.yang (state)					

Green marked describes new capability at IETF 125. "P" to partially implemented.

YANG Message Broker

IETF 125 Implementation Status

	NetGauze	Pmacct	Apache Kafka	Ciena Blueplanet	Cisco Crosswork
draft-ietf-nmop-message-broker-telemetry-message	✓	✓			
YANG-Push Receiver	✓	✓			
YANG Schema Retrieval	✓				
YANG Message Broker Producer	✓				
YANG Schema Registry	na	na	✓	na	na
YANG Message Broker Consumer	✓			(✓)	✓
Validate Telemetry Message against YANG Schema	✓			(✓)	✓
draft-ietf-netmod-yang-anydata-validation	✓				
YANG Schema Comparison	na	na	✓	na	na

NetGauze
Pmacct

Green marked describes new capability at IETF 125. "P" to partially implemented.

Next Steps from IETF 124 Hackathon…

- Yanglint, misses anydata validation support within YANG structures
- Pmacct corrects collection-timestamp to be typedef date-and-time compliant.
- Pmacct corrects encoding value to include ietf-subscribed-notifications namespace in JSON encoding.
- Pmacct corrects network-operator-metadata/labels/anydata-values anydata statement to be valid I-JSON.
- YANG schema registry supporting YANG features
- NetGauze supporting
  - netconf <get> to obtain YANG library content,
  - YANG validation for YANG-Push notifications,
  - YANG schema registration.

Next Steps for IETF 126 and 127 Hackathon…

- Re-do YANG Message Broker Consumer tests on updated testbed with Ciena Blue Planet UAA Workflow Engine
- Complete IETF YANG module subscription tests with 6Wind VSR
- Compare YANG feature coverage between libyang, yangkit and yangtools
- Support draft-ietf-netmod-yang-anydata-validation in yangkit
- Revert import only modules for augmented-by workaround
- Preserve cached YANG-Push subscription state across receiver reload
- Obtain YANG-Push subscription state from publisher if missing
- Support Netconf session queuing for schema retrieval
- Support YANG-Push Message Broker Topic Naming with dynamic Apache Kafka topic creation API

Thanks to…

- Paolo Lucente – Pmacct
- Qin Wu – Huawei
- Michael Mackey – Huawei
- Scott Huan - Cisco
- Rob Wilton – Cisco
- Dan Voyer – Cisco
- Sivakumar Sundaravadivel – Ciena
- IJsbrand Wijnands – Arrcus
- Irfan Mohammad – Arrcus
- Benoit Claise - Everything OPS
- Wanting Du – Swisscom
- Rafael Julio – Swisscom
- Leonardo Rodoni - Swisscom
- Ahmed Elhassany – Swisscom
- Thomas Graf – Swisscom

Want to to know more than join us at…

- NMOP on **Friday 09:00 – 11:00** at the Grand Ballroom 1 for Message Broker, Telemetry Message and Message Key document updates and hackathon related details.